Notes on Dynamic Programming

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[†]I wrote these notes to accompany, but not replace, Jesús Bueren's 2023-2024 lectures for *Macro III - Dynamic Programming and Real Business Cycles* at the EUI. I am the sole responsible for any typos and mistakes. If you spot any or would like to add some comments, please let me know: `josepedro.garcia@eui.eu`.

[**]Typos were corrected and the proof of monotonicity in section 2.3.1 improved.

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1 Introduction

The main purpose of these notes is to provide an introduction to dynamic programming, a fundamental tool of modern macroeconomics.

Most problems in macroeconomics are intrinsically dynamic. How much should we save today in order to consume more tomorrow? Should we decline a job offer today in the hope of having a better one tomorrow? We have seen these problems before and solved them with dynamic optimization methods — typically by relying on a time-0 approach or sequential markets. With dynamic programming we do something different: we use Bellman's principle of optimality to express a large (in)finite problem recursively and thus break it down into what is almost a two-period model. This can be remarkably useful. First, it considerably simplifies the problem in hand. And second, it allows us to address a significant number of models which do not have a closed-form analytical solution. Instead, we can solve those numerically by calling upon the Contraction Mapping Theorem.

The notes were then prepared with two goals in mind. One, to lay out the usefulness and intricate theory of dynamic programming and the Contraction Mapping Theorem. We do this in Section 2 while developing the simplest neoclassical growth model. And two, to demonstrate how those mathematical concepts allow to implement a value function iteration algorithm as a numerical solution method. The latter is carried out computationally, so a set of Matlab codes will also follow attached to these notes. Yet in the last two sections we put in writing what the algorithm is attempting to do. In Section 3 we explore the computational solution for a deterministic neoclassical growth model. In Section 4 we do the same for its stochastic version. As additional material, we also show how value function iteration can be used to solve more complex problems, such as firms' labour demand decisions with non-covexities and an intratemporal choice.

Although these notes follow closely the concepts and notation used by Jesús Bueren in his lectures, **they should be seen as an helpful complement but not as a substitute**, even if an imperfect one. A sizeable portion of the examinable material is not covered in the notes (*e.g.* recursive competitive equilibrium in complete markets), and some of the models developed here are not part of the course (*e.g.* the firms' labour demand model in subsection 4.2.). **When in doubt or in case of potential conflicts, preference should definitely be given to the material provided by Jesús**.

Lastly, the notes were based on material from past studies and a number of references, which should be consulted for a deeper and more technical treatment of the concepts here explored. These include Stokey and Lucas (1989), Adda and Cooper (2003), Sargent and Ljungqvist (2000), and Miao (2014). Additional research material may be cited and duly referenced.

2 The idea behind dynamic programming

The purpose of this section is to gradually build an infinite time horizon optimization problem which can be efficiently solved with dynamic programming, and particularly with numerical methods. This does not mean these are unsuitable tools to deal with finite horizon problems. On the contrary. But as we will rely entirely on the neoclassical growth model, it will be clear that while a finite horizon version has a closed-form analytical solution, the infinite horizon version may not have. This emphasizes the usefulness of setting up the model in recursive form and solving it numerically.

This model can be described as follows. Say we live in an economy wherein a social planner wants to find the optimal paths for consumption and capital, subject to resource and non-negativity constraints. For tractability we will be very specific on the shape of the utility and production functions. Suppose that consumption preferences are given by $u(c_t) = \log(c_t)$ and output is produced only with capital according to $f(k_t) = k_t^\alpha$, where $\alpha \in (0,1)$ stands for the capital share in the economy. Capital depreciates at a rate $\delta \in [0,1]$, and the initial level of capital is given.

In the first subsection we solve this model analytically for an economy which lasts $T < \infty$ periods. Without loss of generality, we conveniently assume that $\delta = 1$ as this makes computations easier. The finite version of the model is shown to have a closed-form analytical solution which depends crucially on the existence of two boundary conditions for capital. In the second subsection, however, one of those conditions is dropped because we move to a model with an infinite time horizon: $T \to \infty$. We will then show how to write the model recursively using Bellman's principle of optimality. This can be extraordinarily useful because the Contraction Mapping Theorem ensures that a solution to such a problem does exist and can be found with consecutive iterations. That is what we do in the third subsection: we start by diving deep into the mathematics of the Contraction Mapping Theorem and then show that its implications can be used to solve the infinite horizon model numerically.

2.1 Finite horizon problem

We start with the neoclassical growth model in finite time. Using the functions described above, the problem of the social planner of the economy can be formally expressed as follows:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \left\{ \sum_{t=0}^{T} \beta^t \log(c_t) \right\} \quad \text{subject to} \quad c_t + k_{t+1} = k_t^\alpha + (1-\delta)k_t \;\wedge\; c_t \geq 0 \;\wedge\; k_0 \text{ is given} \tag{1}$$

The planner discounts lifetime utility according to $\beta \in (0,1)$. Notice that capital is pre-determined at the start of each period. In other words, this means that the stock of capital in period t was defined in $t-1$. Upon waking up the planner knows immediately the amount of existing resources in the economy,

$k_t^\alpha + (1-\delta)k_t$. His problem is then to optimally decide on how much to consume (c_t) and how much to save for next period (k_{t+1}). Naturally there is only one degree of freedom: the consumption decision pins down the capital decision and vice-versa, as it is clear from the resource constraint, $c_t + k_{t+1} = k_t^\alpha + (1-\delta)k_t$. For tractability, we can drop the second component of the right-hand side. Basically, let there be full depreciation of capital after one period, such that $\delta = 1$. This just makes computations easier and does not alter the main argument we will build. [1]

The resource constraint can be used to rewrite consumption as a function of capital. To find an interior solution we set up the following lagrangian:

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t \log\left(k_t^\alpha - k_{t+1}\right)$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 \iff -\beta^t \frac{1}{k_t^\alpha - k_{t+1}} + \beta^{t+1} \frac{\alpha k_{t+1}^{\alpha-1}}{k_{t+1}^\alpha - k_{t+2}} = 0 \iff \beta \frac{\alpha k_{t+1}^{\alpha-1}}{k_{t+1}^\alpha - k_{t+2}} = \frac{1}{k_t^\alpha - k_{t+1}}$$

As shown above, the solution to the problem is a second-order difference equation. We thus need two (boundary) conditions, which we do have. First, there is an exogenously given initial level of capital k_0 — it is a constraint in (1). That value is assumed to be known, so we can indeed use it as a boundary condition. And second, since this is a finite problem, the strict increasingness of the utility function ensures that $k_{T+1} = 0$. In other words, at the end of time there cannot be any resources left, otherwise the social planner could do better by allocating those to consumption before the economy shuts down. We will show that these two conditions are both necessary and sufficient to pin down the optimal sequence of capital.

We start by rewriting the first-order condition:

$$\beta \frac{\alpha k_{t+1}^{\alpha-1}}{k_{t+1}^\alpha - k_{t+2}} = \frac{1}{k_t^\alpha - k_{t+1}} \iff \alpha\beta k_{t+1}^{\alpha-1}\left(k_t^\alpha - k_{t+1}\right) = k_{t+1}^\alpha - k_{t+2} \iff \alpha\beta\left(\frac{k_t^\alpha}{k_{t+1}} - 1\right) = 1 - \frac{k_{t+2}}{k_{t+1}^\alpha}$$

Nothing substantial has changed: we still have a second-order difference equation. Yet let us define the variable $z_t \equiv k_{t+1}/k_t^\alpha$, for any period t. Then, the solution actually becomes a first-order difference equation:

$$\alpha\beta\left(\frac{1}{z_t} - 1\right) = 1 - z_{t+1} \tag{2}$$

With a first-order difference equation we only need to rely on a single boundary condition. So, let us take the terminal condition, $k_{T+1} = 0$, and solve the equation backwards:

$z_T = \frac{k_{T+1}^\alpha}{k_t^\alpha} = \frac{0}{k_t^\alpha} = 0;$

$1 - z_T = \alpha\beta\left(\frac{1}{z_{T-1}} - 1\right) \iff \frac{1}{z_{T-1}} = \frac{1}{\alpha\beta} + 1 \iff z_{T-1} = \frac{\alpha\beta}{1+\alpha\beta};$

[1] In the infinite version of the model, the chosen functional forms and the assumption that $\delta = 1$ also enable a closed-form analytical solution. This is known as the Brock & Mirman model.

$1 - z_{T-1} = \alpha\beta\left(\frac{1}{z_{T-2}} - 1\right) \iff \frac{1}{z_{T-2}} = \frac{1}{\alpha\beta(1+\alpha\beta)} + 1 \iff z_{T-2} = \frac{\alpha\beta+(\alpha\beta)^2}{1+\alpha\beta+(\alpha\beta)^2};$

$1 - z_{T-2} = \alpha\beta\left(\frac{1}{z_{T-3}} - 1\right) \iff \frac{1}{z_{T-3}} = \frac{1}{\alpha\beta\left(1+\alpha\beta+(\alpha\beta)^2\right)} + 1 \iff z_{T-3} = \frac{\alpha\beta+(\alpha\beta)^2+(\alpha\beta)^3}{1+\alpha\beta+(\beta)^2+(\alpha\beta)^3};$

$\vdots$

Clearly a pattern emerges. We can then solve for a generic element z_t:

$$\begin{aligned} z_{T-\tau} = \frac{\alpha\beta + ... + (\alpha\beta)^{\tau}}{1 + \alpha\beta + ... + (\alpha\beta)^{\tau}} \implies z_t &= \frac{\alpha\beta + ... + (\alpha\beta)^{T-t}}{1 + \alpha\beta + ... + (\alpha\beta)^{T-t}} = \alpha\beta \cdot \frac{1 + ... + (\alpha\beta)^{T-t-1}}{1 + \alpha\beta + ... + (\alpha\beta)^{T-t}} \\ &= \alpha\beta \cdot \frac{1 - (\alpha\beta)^{T-t}}{1 - \alpha\beta} \cdot \left(\frac{1 - (\alpha\beta)^{T-t+1}}{1 - \alpha\beta}\right)^{-1} \\ &= \alpha\beta \cdot \frac{1 - (\alpha\beta)^{T-t}}{1 - (\alpha\beta)^{T-t+1}} \end{aligned}$$

But recall that $z_t \equiv k_{t+1}/k_t^{\alpha}$. Using this definition, we can then express the optimal choice of capital as a first-order difference equation instead of the second-order difference equation given by the FOC.

$$k_{t+1} = \alpha\beta \cdot \frac{1 - (\alpha\beta)^{T-t}}{1 - (\alpha\beta)^{T-t+1}} \cdot k_t^{\alpha} \tag{3}$$

The equation above is interesting for two reasons. First, it shows that the optimal choice of capital (k_{t+1}, the left-hand side) is only a function of capital chosen in the previous period and carried over to the present (k_t on the right-hand side). The solution to the problem in (1) must then be a sequence generated by **policy function** of the type $k_{t+1} \equiv \phi(k_t)$: for every possible value of capital which may have already been accumulated, $\phi(\cdot)$ prescribes exactly how much capital is to be saved today. And second, notice that for particular parameters of α and β there is an infinite combination of $\{k_t, k_{t+1}\}_{t=0}^{T}$ which satisfies equation (3) above. We then need another boundary condition to pin down an exact solution.

Here we can rely on the initial condition for capital. Since k_0 is given we can iterate forward from $t = 0$ on in order to obtain the choices of capital in each period.

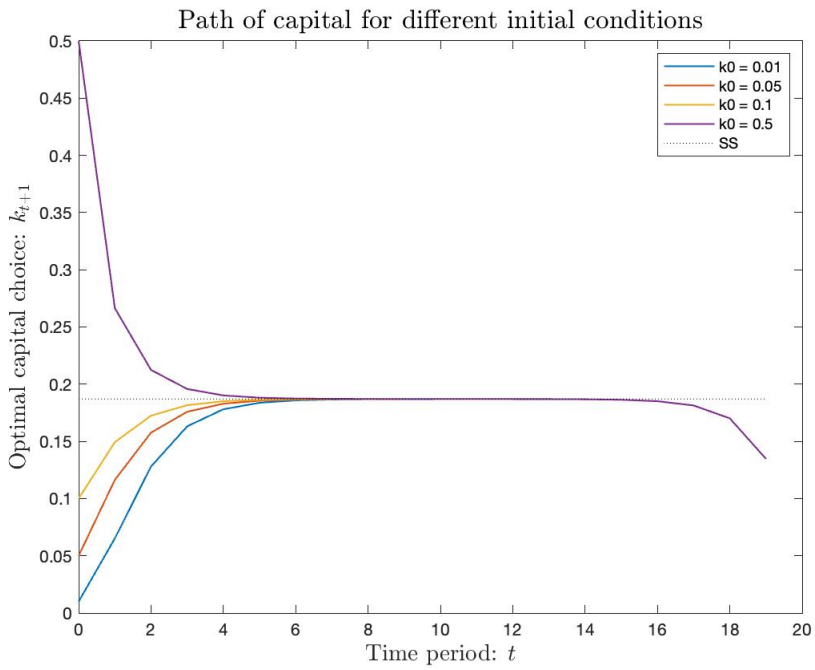
$k_1 = \alpha\beta \cdot \frac{1-(\alpha\beta)^T}{1-(\alpha\beta)^{T+1}} \cdot k_0^{\alpha};$

$k_2 = \alpha\beta \cdot \frac{1-(\alpha\beta)^{T-1}}{1-(\alpha\beta)^T} \cdot k_1^{\alpha} = (\alpha\beta)^{1+\alpha} \cdot \frac{1-(\alpha\beta)^{T-1}}{1-(\alpha\beta)^T} \cdot \left(\frac{1-(\alpha\beta)^T}{1-(\alpha\beta)^{T+1}}\right)^{\alpha} \cdot k_0^{\alpha^2};$

$\vdots$

The initial value of capital pins down the entire sequence throughout time. As such, we can in fact express the policy function of the model as $k_{t+1} \equiv \phi(k_0)$, for any $t = \{0, 1, ..., T-1\}$. Figure 1 below emphasizes this point. For different values of k_0, we rely on the policy function expressed in (3) to demonstrate that yes, optimal sequences do differ, but they eventually converge to a steady-state level of capital.

Figure 1: Model solution with different values of k_0



Notes: Simulations for $T = 20$ periods with $\alpha = 0.36$ and $\beta = 0.95$. The steady-state (SS) level of capital is given by $k = (\alpha\beta)^{1/(1-\alpha)}$.

2.2 Infinite horizon problem

To solve the previous problem in a *finite* horizon we relied on two boundary conditions: first, the economy is somehow endowed with an initial level of capital, k_0, whose value we do know; and second, a terminal condition which posits that $k_{T+1} = 0$. So suppose now we want to solve the exact same problem as before but in an *infinite* time horizon: $t = \{0, 1, ...\}$.

This creates an immediate problem: setting up the problem as in (1) will yield the same second-order difference equation but one boundary condition is lost. More specifically, the terminal condition no longer applies precisely because there is no more a terminal period of time. Importantly, the initial condition still holds: we do know the value of k_0.

In order to circumvent this issue let us try to work around the problem. More specifically, consider the following function:

$$V(k_0) = \max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \log(c_t) \right\} \quad \text{subject to} \quad c_t + k_{t+1} = k_t^{\alpha} \;\; \wedge \;\; c_t \geq 0 \tag{4}$$

For each initial value of capital, $V(\cdot)$ is a function which yields the present discounted utility value the social planner draws from optimally solving this constrained maximization problem. As we saw in the previous subsection, it is the initial condition for k_0 which pins down the exact *levels* of capital and consumption

chosen by the social planner. Therefore, as the level of k_0 varies so too does the utility value drawn by the social planner at the optimal solution. We call $V(k_0)$ a **value function**.

That value function can actually be expanded. As we plug the resource constraint in the utility function and open the summation:

$$\begin{aligned} V(k_0) &= \max_{\{k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \log(k_t^\alpha - k_{t+1}) \right\} = \max_{\{k_{t+1}\}_{t=0}^{\infty}} \left\{ \log(k_0^\alpha - k_1) + \beta(k_1^\alpha - k_2) + \beta^2(k_2^\alpha - k_3) + ... \right\} \\ &= \max_{\{k_{t+1}\}_{t=0}^{\infty}} \left\{ \log(k_0^\alpha - k_1) + \sum_{t=1}^{\infty} \beta^t \log(k_t^\alpha - k_{t+1}) \right\} \\ &= \max_{\{k_{t+1}\}_{t=0}^{\infty}} \left\{ \log(k_0^\alpha - k_1) + \beta \sum_{t=1}^{\infty} \beta^{t-1} \log(k_t^\alpha - k_{t+1}) \right\} \end{aligned}$$

For convenience only, in this last step we pulled the discount factor β out of the sum. It will become clear why in the following steps. Notice that the value function is the maximum of a sum of two components: the utility level drawn in period 0 and the present discounted utility derived from $t = 1$ on, which is also known as the discounted continuation value. Here we can make use of Bellman's principle of optimality: *an individual following an optimal strategy from $t + 1$ on, can do no better than to optimize at time t taking future optimal plans as given*.[2] Under this principle we can further expand the value function as follows:

$$V(k_0) = \max_{k_1} \left\{ \log(k_0^\alpha - k_1) + \beta \max_{\{k_{t+1}\}_{t=1}^{\infty}} \left\{ \sum_{t=1}^{\infty} \beta^{t-1} \log(k_t^\alpha - k_{t+1}) \right\} \right\} \tag{5}$$

The social planner is looking for a path of infinite levels of capital, $\{k_{t+1}\}_{t=0}^{\infty}$, taking k_0 as given. Bellman's principle of optimality simply states that the social planner can just look for the optimal level of capital to be saved in $t = 0$, while taking as given the optimal choices he would make from $t = 1$ on as a result of his initial decision. This is precisely what is formally prescribed by the equation (5) above.

But let us focus precisely on those optimal future choices. The second max operator seems to resemble something we have already seen:

$$\begin{aligned} \max_{\{k_{t+1}\}_{t=1}^{\infty}} \left\{ \sum_{t=1}^{\infty} \beta^{t-1} \log(k_t^\alpha - k_{t+1}) \right\} &= \max_{\{k_{t+1}\}_{t=1}^{\infty}} \left\{ \log(k_1^\alpha - k_2) + \beta \log(k_2^\alpha - k_3) + \beta^2 \log(k_3^\alpha - k_4) + ... \right\} \\ &= V(k_1) \end{aligned}$$

This is just the value function of starting the problem with an initial level of capital k_1, exactly the value which is optimally chosen in $t = 0$ and carried to period $t = 1$.

We started with the definition of a value function and by relying on Bellman's principle of optimality we can ultimately achieve the following **Bellman equation**:

$$V(k_0) = \max_{k_1} \left\{ \log(k_0^\alpha - k_1) + \beta V(k_1) \right\} \tag{6}$$

[2] The original statement is "*An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.*" (see Bellman (1957), ch. 3)

The Bellman equation displayed in (6) sets up the exact same problem as in (4). Yet it does so in a more concise expression which builds upon Bellman's optimality principle. The infinite horizon problem is thus broken down into an almost two-period model: the social planner chooses optimal capital today (say, in period $t = 0$ he sets k_1) while assuming that with that initial choice he will remain behaving optimally from tomorrow on. This is the core of dynamic programming.

2.2.1 Solution with the recursive formulation

The Bellman equation allows to write the maximization problems *recursively*, that is taking into account Bellman's principle of optimality. Two additional remarks must also be made. First, equation (6) above sets up the problem for a social planner who begins his choice in period $t = 0$ and with a known level of capital k_0. But we can start from wherever we want. Suppose that the social planner wakes up on a random day and knows he has exactly k units of capital. This is his **state variable**: it fully describes everything the social planner needs to know about the economy (namely its existing resources) in order to make an informed decision about his choices that particular day. And which choices are those? In this case only one: he wants to choose capital to carry for the next period, k'. This is his **choice/control variable**.

Second, notice that the choice of capital is not unconstrained. On the one hand, it would not make sense to consider negative levels of capital: $k' \geq 0$. On the other, since there is a non-negativity constraint on consumption too, the choice of capital must also be bounded from above: $c = k^\alpha - k' \geq 0 \implies k' \leq k^\alpha$. The set of admissible values for k' must then be $\Gamma(k) \equiv [0, k^\alpha]$. We call $\Gamma(k)$ the **feasible set** of values for the choice variable of the social planner, which depends on the state variable.

We can then finally rewrite the Bellman equation as follows:

$$V(k) = \max_{k' \in \Gamma(k)} \left\{ \log\left(k^\alpha - k'\right) + \beta V(k') \right\} \tag{7}$$

An interior solution to the problem must necessarily satisfy the first-order condition:

$$\text{FOC wrt } k' : \quad -\frac{1}{k^\alpha - k'} + \beta V_1(k') = 0,$$

where $V_1(\cdot)$ stands for the derivative of the value function with respect to its first (and only) argument. But if we do not know the shape of the value function, how can we possibly know the shape of its first derivative?

Here we rely on two additional concepts. First, suppose we know the **policy function** for capital holdings: $k' \equiv \phi(k)$. This is the object the social planner wants to find: a rule which tells him how much should he save now for each possible level of capital carried from the previous period. This is basically the decision rule the social planner must follow in order to optimally solve the problem. [3] It is, in other words, the

[3] In the finite horizon version of the model this prescription was given in equation (3).

argument which maximizes the Bellman equation above.

$$k' \equiv \phi(k) = \arg\max_{k' \in \Gamma(k)} \left\{ \log\left(k^\alpha - k'\right) + \beta V(k') \right\} \implies V(k) = \log\left(k^\alpha - \phi(k)\right) + \beta V\left(\phi(k)\right)$$

Second, we can use that policy function to also work with the **envelope condition**. Taking the derivative of the value function with respect to k:

$$\begin{aligned}
V_1(k) &= \frac{\alpha k^{\alpha-1} - \phi_1(k)}{k^\alpha - \phi(k)} + \beta V_1\left(\phi(k)\right) \cdot \phi_1(k), \quad \text{where } \phi_1(k) \text{ is the first derivative of the policy function wrt } k \\
&= \frac{\alpha k^{\alpha-1}}{k^\alpha - \phi(k)} + \phi_1(k) \cdot \underbrace{\left(-\frac{1}{k^\alpha - \phi(k)} + \beta V_1\left(\phi(k)\right) \right)}_{= 0 \text{ by the FOC}} \\
&= \frac{\alpha k^{\alpha-1}}{k^\alpha - \phi(k)} = \frac{\alpha k^{\alpha-1}}{k^\alpha - k'} \\
&= \frac{\partial}{\partial k}\left(\log(k^\alpha - k') \right)
\end{aligned}$$

The envelope condition yields the derivative of the value function, even when the shape of the latter remains unknown. So, since $V_1(k) = \alpha k^{\alpha-1}/(k^\alpha - k')$, we can also claim that $V_1(k') = \alpha k'^{\alpha-1}/(k'^\alpha - k'')$, where k'' is the level of capital two periods from now. Or, in other words, the level of capital the social planner will optimally choose tomorrow. The FOC then becomes as follows:

$$-\frac{1}{k^\alpha - k'} + \beta \cdot \frac{\alpha k'^{\alpha-1}}{k'^\alpha - k''} = 0 \iff \beta \cdot \frac{\alpha k'^{\alpha-1}}{k'^\alpha - k''} = \frac{1}{k^\alpha - k'}$$

This is exactly the same second-order difference equation we had obtained in the finite time horizon problem. It makes sense: other than the time horizon faced by the planner, which can be extremely important, nothing else has changed. The condition for optimality is therefore the same.

When we dealt with a finite horizon problem the existence of an optimal capital sequence was pinned down by two boundary conditions. We used these to iterate on the optimal condition provided by the FOC. However, once we moved to an infinite time horizon problem one of those conditions was immediately dropped: there cannot be a terminal value of capital because time is infinite. While relying on Bellman's principle of optimality we were able to rewrite the problem in a simpler and more concise fashion. But given that a terminal condition is still missing, are there any material gains of doing so?

As a matter of fact there are. Increasingly complex problems in modern macroeconomics will rarely have closed-form analytical solutions as the one we found in the previous subsection. We can, however, accurately approximate a solution via numerical methods. One such method is value function iteration, which we will study extensively in the following sections. Yet to implement it we must first express our problem recursively, as in equation (7), and then ensure that the Contraction Mapping Theorem holds. In the next section we take a mathematical detour to explore this powerful theorem. We will also show that

a solution to the neoclassical growth model can indeed be found with value function iteration because the Bellman equation displayed in (7) defines a contraction mapping.

2.3 Contraction Mapping Theorem

This subsection is purely mathematical. We will first define a contraction mapping and then go through the implications postulated by the Contraction Mapping Theorem. We will show that these can be remarkably useful to find a solution to the model we have been developing so far.

Start by considering a metric space (S, d), where S is a non-empty set, and $d : S \times S \to \mathbb{R}$ a function which sets the distance between any two elements of S. This distance function must satisfy the following three conditions[4]:

- Non-negative distances: $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$;
- Symmetry of distances: $d(x, y) = d(y, x)$;
- Triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

The following two cases are then examples of metric spaces.

Example 1. Consider the set $S = \mathbb{R}^2$ whose generic element is $x \equiv (x_1, x_2)$. The Euclidean distance between two points x and y corresponds to the following:

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

The duple formed by the set $\mathbb{R}^2$ and the Euclidean distance above is a metric space.

Example 2. Let $S = \mathfrak{B}(X)$ be the set of bounded functions whose domain is a non-empty set X. Formally, a generic element of S is a function $f : X \to \mathbb{R}$. The sup distance between any two bounded functions f_1 and f_2 in $\mathfrak{B}(X)$ corresponds to the following:

$$d(f_1, f_2) = \sup_{x \in X} |f_1(x) - f_2(x)|,$$

where boundedness implies that $\sup_{x \in X} |f(x)| < \infty$ for any $f \in \mathfrak{B}(X)$. This duple is also a metric space.

Consider now an operator $T : S \to S$. Basically, let T be a generic operation which maps elements from S to an element also in S. Take the second example above. We can think of an operator T as the following: $Tf(x) = 0.5 \times f(x)$. In this very simple case, T is just an operator which takes any bounded function $f(x) \in \mathfrak{B}(X)$ as an input and spits out yet another bounded function: call it $g(x) = 0.5 \times f(x)$.

We are mostly interest in operators which can be defined as a contraction mapping.

[4] These three properties, and specially the third, are useful to better understand the proof of the Contraction Mapping Theorem.

Definition 1. ***(Contraction mapping).*** *Let (S, d) be a metric space, and let $T : S \to S$ be an operator. We say that T is a contraction mapping with modulus β if for some $\beta \in (0, 1)$ the following holds:*

$$d\big(Tx, Ty\big) \leq \beta\, d(x, y), \qquad \forall\, x, y \in S$$

Say that x and y are two arbitrary elements of the metric space (S, d). These can be simple points in $\mathbb{R}^2$ as in the first example, or two bounded functions as in the second. According to the definition above, T is a contraction if the distance between these two elements shrinks by a factor of $\beta \in (0, 1)$ every time the operator is applied. In other words, applying T to x and y brings them closer to each other.

This can be an interesting property if we do know that T is a contraction mapping. But how can we be sure? Set S can be infinitely large for us to be computing distances between any two elements of that set. In addition, the theorem holds if we can find any value of β between 0 and 1 which produces such inequality. But that range also comprises an infinite number of elements, which can again make things admittedly complicated.

To circumvent these issues we can instead rely on *Blackwell's sufficient conditions*. David Blackwell was a mathematician who demonstrated that two conditions alone are sufficient, though not necessary, for an operator $T : S \to S$ to be a contraction mapping. Let again $S = \mathfrak{B}(X)$ be the set of all bounded functions with domain in a non-empty set X. Blackwell's two sufficient conditions are the following:

1. **Monotonicity**. Say that f and g are two bounded functions with domain in X: $f, g \in S$. Suppose also that $f(x) \leq g(x)$, for all $x \in X$. The monotonicity condition implies that applying the operator T should always preserve that ranking: $Tf(x) \leq Tg(x)$.

2. **Discounting**. Consider again a bounded function f with domain in X, such that $f \in S$. For any constant $a \in \mathbb{R}_+$, there must exist a value $\beta \in (0, 1)$ which ensures the following:

$$T\big(f(x) + a\big) \leq Tf(x) + \beta a$$

If the two Blackwell conditions hold, then T is shown to be a contraction mapping with modulus $\beta \in (0, 1)$. At the end of this subsection we show how to rely on these to prove that the recursive formulation in equation (7) is indeed a contraction mapping.

For the time being we take that as given: the Bellman equation associated with the growth problem is a contraction mapping. From **Definition 1** we know exactly what that means. But why is that useful to find a solution to the problem? This is where the **Contraction Mapping Theorem (CMT)** comes in handy.

Theorem 1. ***(Contraction Mapping Theorem).*** *Say that (S, d) is a complete metric space and $T : S \to S$ a contraction mapping with modulus β. Then we know the following is true:*

1. *The operator T has a unique fixed point $x^* \in S$, such that applying T to x^* yields back x^*: $Tx^* = x^*$.*

2. *For any $x_0 \in S$, $d\left(T^i(x_0), x^*\right) \leq \beta^i d(x_0, x^*)$, for all integers $i = \{0, 1, 2, ...\}$.*

Proof. See Appendix A. □

Let x_0 be a generic element of S. We can define a sequence whereby we iteratively apply the operator T starting from x_0. In the first iteration we obtain $x_1 = Tx_0$, in the second $x_2 = Tx_1$, in the third $x_3 = Tx_2$, and from then on. The CMT offers two important results about this sequence. It postulates that, regardless of the starting point, the sequence generated by iterative applications of T converges to a point x^* which is also an element of S (point 2 of the theorem). And, just as importantly, it states that x^* is actually a unique fixed point of T (point 1 of the theorem).

So far we have kept the notation as generic as possible. Going back to the two examples above, S can either be $\mathbb{R}^2$ or even the set of all bounded functions with a particular domain, $\mathfrak{B}(\cdot)$. In this latter case the fixed point x^* would actually be a bounded function: applying T over that function x^* would simply return x^* again.

This is a particularly adequate example in light of the problem we have in hand. In the next subsection we show that the recursive formulation of the neoclassical growth model we have been developing does define a contraction mapping. We demonstrate this with Blackwell's sufficient conditions. From the CMT this must mean that there exists a value function which is a fixed point of the problem, and which we can find through a finite number of iterative applications of the defined operator.

2.3.1 Is our recursive formulation a contraction mapping?

Recall the growth problem as defined in equation (7):

$$V(k) = \max_{k' \in \Gamma(k)} \left\{ \log\left(k^\alpha - k'\right) + \beta V(k') \right\}$$

Focus on the right-hand side. It takes a function $V : \mathbb{R}_+ \to \mathbb{R}$ as an input and applies to it a certain operation: that function $V(\cdot)$ is multiplied by a constant, added to another function and then the corresponding max is computed. This can be done for every point of the domain of $V(\cdot)$. In return, that right-hand side spits out a new function: that of the left-hand side.

We can then iteratively generate new functions. Start by plugging an arbitrary $V_0(k)$ function on the right-hand side. We can generate a $V_1(k)$ by applying the operation defined in (7) to that initial function. And then we can also generate $V_2(k)$ from $V_1(k)$. And then $V_3(k)$ from $V_2(k)$. We can actually proceed

indefinitely, but the pattern is clear:

$$
\begin{aligned}
&V_{i+1}(k) = \max_{k' \in \Gamma(k)} \left\{ \log\left(k^{\alpha} - k'\right) + \beta V_i(k') \right\}, \quad \forall\, i = \{0, 1, ...\} \\
&\implies V_{i+1} = TV_i
\end{aligned}
\tag{8}
$$

From the CMT we know that if the operator T in equation (8) is a contraction mapping then i) there exists a unique value function V^* which is a fixed point of this operator, $V^* = TV^*$, and *ii)* we inevitably reach that unique fixed point by continuously iterating on whatever initial guess we may have for the value function. All we have to do then is to show that (8) is a contraction mapping by satisfying Blackwell's sufficient conditions.

Monotonicity:

Consider two arbitrary functions defined on the set of non-negative real numbers: $W : \mathbb{R}_+ \to \mathbb{R}$ and similarly $Q : \mathbb{R}_+ \to \mathbb{R}$. Say also that $W(k) \geq Q(k)$, for all $k \in \mathbb{R}_+$. Under this assumption, the monotonicity condition requires that $TW(k) \geq TQ(k)$ along the entire domain $\mathbb{R}_+$.

We start by acknowledging that each of these two value functions define their one policy function. For example, for $Q(\cdot)$ the policy function corresponds to the following:

$$
\phi_Q(k) = \arg\max_{k' \in \Gamma(k)} \left\{ \log(k^{\alpha} - k') + \beta Q(k') \right\}
$$

and similarly for $\phi_W(\cdot)$. Using these, we can then set up the following set of relationships:

$$
\begin{aligned}
TQ(k) &= \max_{k' \in \Gamma(k)} \left\{ \log(k^{\alpha} - k') + \beta Q(k') \right\} = \log\left(k^{\alpha} - \phi_Q(k)\right) + \beta Q\left(\phi_Q(k)\right) \\
&\leq \log\left(k^{\alpha} - \phi_Q(k)\right) + \beta W\left(\phi_Q(k)\right) && \text{because } \beta Q(\cdot) \leq \beta W(\cdot) \\
&\leq \log\left(k^{\alpha} - \phi_W(k)\right) + \beta W\left(\phi_W(k)\right) && \text{by definition of } \phi_W(\cdot) \text{ as arg max of } W(\cdot) \\
&= \max_{k' \in \Gamma(k)} \left\{ \log(k^{\alpha} - k') + \beta W(k') \right\} = TW(k)
\end{aligned}
$$

Therefore, we do have that $Q(k) \leq W(k) \implies TW(k) \geq TQ(k)$. This means that the monotonicity condition is satisfied. □

Discounting:

Consider a constant real value $a \in \mathbb{R}_+$ and the usual value function $V : \mathbb{R}_+ \to \mathbb{R}$. The discounting property

holds if there exists some $\beta \in (0,1)$ for which $T\big(V(k)+a\big) \leq TV(k) + \beta a$.

$$
\begin{aligned}
T\big(V(k)+a\big) &= \max_{k' \in \Gamma(k)} \Big\{ \log\left(k^{\alpha} - k'\right) + \beta\big(V(k')+a\big) \Big\} && \text{by definition} \\
&= \max_{k' \in \Gamma(k)} \Big\{ \log\left(k^{\alpha} - k'\right) + \beta V(k') + \beta a \Big\} && \text{by distributive property} \\
&= \max_{k' \in \Gamma(k)} \Big\{ \log\left(k^{\alpha} - k'\right) + \beta V(k') \Big\} + \beta a && \beta a \text{ is a constant} \\
&= TV(k) + \beta a
\end{aligned}
$$

The model is parametrized with a discount factor β which lies between 0 and 1. Therefore, the discounting property binds with equality for any $\beta \in (0,1)$. The discounting property is also satisfied. □

2.3.2 Value function iteration as a numerical solution method

In the previous subsection we showed that the problem in (7) is a contraction mapping because it satisfies Blackwell's sufficient conditions. Therefore, we can apply the CMT. This immediately tells us two things. First, there is a unique value function $V^*(k)$ which is a fixed point of the problem in (7). This does not mean that $V(k) = k$, for all $k \in \mathbb{R}_+$. Instead, it means that there exists a function upon which the operation (8) is applied and the outcome is that same function. And second, we do know that continuous iterations of that operator on any initial guess of the value function will inevitably converge to that unique fixed point.

This is the goal of a value function iteration algorithm. We do not solve for the analytical expression of the value function, but instead rely on the existence, uniqueness and convergence properties of the CMT to eventually find it numerically (an approximation of its behaviour for certain levels of the domain). In the next section we will demonstrate how this procedure can be computationally and efficiently implemented.

3 Deterministic dynamic programming problems

The purpose of this section is to solve the infinite time neoclassical growth model using dynamic programming and a value function iteration algorithm. We can do so because we already wrote the model recursively, and showed that the resulting Bellman equation defines a contraction mapping. This ensures that, regardless of our starting point, we can iterate on the value function until we inevitably achieve a unique fixed point. Although the description of the intended algorithm is self-contained, it replicates what is built and produced in the Matlab script attached to these notes.

There is a single subsection, where we revisit the model and explain what the value function iteration procedure attempts to do. Yet there are two remarks which must be raised. First, we introduce a nuance to the model. For tractability we have been working with a full depreciation rate: $\delta = 1$. Capital is saved

in period t, it serves the purpose of creating additional output in $t+1$ and after that it vanishes. In order to have a model even closer to that of the lectures and the problem sets, suppose now that $\delta \in (0,1)$. We basically introduce the generic resource constraint in the Bellman equation[5], which obviously remains a contraction mapping.

Second, we must now explicitly address an assumption which will be challenged in the following section. Notice that there is no source of stochasticity in the economy. This is important because it simplifies the discounted value of future optimal choices of the planner, $\beta V(k')$. By choosing k' today, the deterministic setting of the economy allows the planner to know *ex-ante* the value he draws from making yet another optimal choice tomorrow with k' units of capital.

3.1 Neoclassical growth model without uncertainty

We start with the nuanced version of the infinite time horizon model we have been developing so far. With $\delta \in (0,1)$ and the same functional forms for utility and production, the social planner of the economy wants to find a policy function for capital which solves the following problem:

$$V(k) = \max_{k' \in \Gamma(k)} \left\{ \log\left(k^{\alpha} + (1-\delta)k - k'\right) + \beta V(k') \right\}, \tag{9}$$

where the feasible set of the choice variable is now $\Gamma(k) \equiv [0, k^{\alpha} + (1-\delta)k]$. Our goal is to retrieve two main objects of interest: *i)* the policy function $k' \equiv \phi(k)$ which prescribes the optimal amount of capital to be saved each period, and *ii)* the value function $V(k)$ which yields the discounted utility value drawn from optimally solving the problem. To do so we now rely on the properties of a contraction mapping. On the one hand, we know that there must exist a value function which is a unique fixed point of the problem in (9). On the other, we also know that by starting with whatever guess for the value function, a finite number of iterations will inevitably lead us to that fixed point. To implement this procedure we first discretize the state and choice vectors of the planner (which in this case coincide as they comprise the same single variable, capital), so that we can eventually build an algorithm which evaluates consecutive iterations of a value function at different values of state and chosen capital.

3.1.1 Discretize the state space: capital

Take the value function defined in (9). Since capital must intuitively take non-negative values only, we can define V as a function which maps elements from $\mathbb{R}_+$ to $\mathbb{R}$. The domain is then a continuum: it can take infinitely many non-negative values. To circumvent this we *discretize* the state space. In other words, we

[5] See equation (1).

define a considerably large set of values for capital (the only state variable) which we can rely on to evaluate the value functions. We call this a **grid** for capital.

That grid can obviously contain whatever non-negative values we may want. Yet a standard practice is to set it up around the steady-state level of capital. From the FOC and the envelope condition:

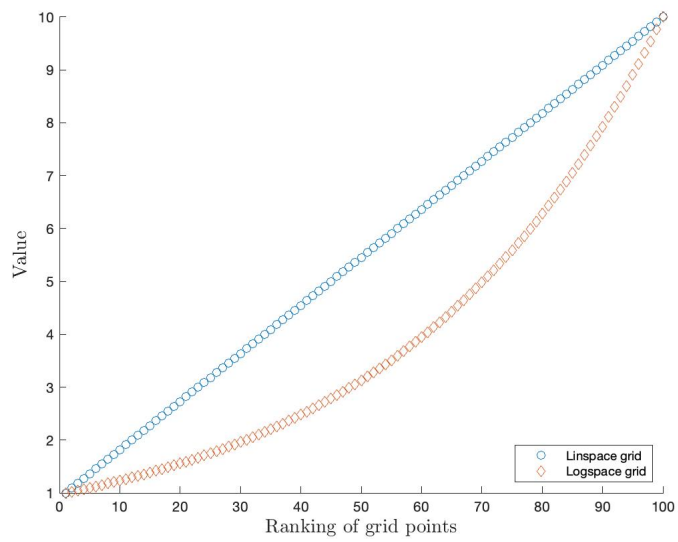
$$-\frac{1}{k^\alpha + (1-\delta)k - k'} + \beta V_1(k') = 0 \qquad \text{from the FOC}$$

$$\iff -\frac{1}{k^\alpha + (1-\delta)k - k'} + \beta \cdot \frac{\alpha k'^{\alpha-1} + (1-\delta)}{k'^\alpha + (1-\delta)k' - k''} = 0 \qquad \text{from the envelope condition}$$

$$\text{In steady-state} \implies -\frac{1}{k^\alpha + (1-\delta)k - k} + \beta \cdot \frac{\alpha k^{\alpha-1} + (1-\delta)}{k^\alpha + (1-\delta)k - k} = 0 \iff k = \left(\frac{\alpha\beta}{1-\beta(1-\delta)}\right)^{\frac{1}{1-\alpha}}$$

In Problem Set 2 we create a grid whose lower bound is 90% of this steady-state level, whereas the upper bound is 110%. The Matlab command for an equally spaced grid is `linspace`. Depending on the nature of the problem we may instead want to have a grid which is very fine on the left end and coarser on the right. The Matlab command for that is `logspace`. Figure 2 below exemplifies the difference between the two alternatives when we set up a grid of 100 points between 0 and 10.

Figure 2: Equally spaced and logarithmic grids



Following the more standard practice, suppose we choose an equally spaced grid. The only question remaining is then to settle its size. To remain as general as possible say we set it up with `nkk` points. The actual choice of `nkk` is governed by a trade-off. On the one hand, we want it to be as large as possible. As we evaluate the value function in an increasing number of points/values, the more accurate our solution can be. On the other, a rising grid size comes at a computational cost: it takes more time for the fixed point to be

found precisely because we are requesting more computations. This can be particularly troublesome for at least two reasons. One, we may be interested in solving the problem as quickly and efficiently as possible. If, for example, the algorithm is part of a simulated method of moments exercise, it will presumably be run dozens of times under different parameters. Computational efficiency may be a priority in that case. And two, the trade-off becomes strikingly clear as the size of the state vector increases. Below we show that solving this deterministic model with a capital grid of `nkk` $= 500$ points takes less than a quarter of a second for the algorithm to find the fixed point. But in the next section we will also show that as stochastic productivity is introduced in a grid of only `nzz` $= 20$ points, both the number of required computations ($500 \times 20 = 10\,000$) and computational time (5.4 seconds) rise exponentially.

3.1.2 VFI algorithm

We want to solve the model in (9) via value function iteration with an equally spaced grid for capital of `nkk` points. The goal is to numerically approximate the value function $V(\cdot)$, which is a fixed point to the problem, and from there to retrieve a policy function for k'.

We kick off the iterative process with a simple guess. The CMT ensures that this initial guess is completely irrelevant: wherever we start we will inevitably end up at the fixed point we are looking for. To ease, let us say that the value function which starts the iteration (call it V_0) takes the value 0 at every single level of capital in our grid: $V_0(k) = 0$, for all k in the grid. Using the sequence generated by the operator, as in (8), the Bellman equation gives us the updated value function:

$$V_1(k) = \max_{k' \in \Gamma(k)} \left\{ \log\left(k^\alpha + (1-\delta)k - k'\right) + \beta V_0(k') \right\} = \max_{k' \in \Gamma(k)} \left\{ \log\left(k^\alpha + (1-\delta)k - k'\right) \right\}$$

To understand what the algorithm does we start by setting a value for the state variable: $\bar{k}$. This is only one of `nkk` potential values of state capital in the grid. And for each level that we may fix, there are also `nkk` possible choices of k'. The one to be selected is then the one which yields the greatest value in the following $1 \times$ `nkk` vector:

$$V_1(\bar{k}) = \max \left\{ \left[\log\left(\bar{k}^\alpha + (1-\delta)\bar{k} - k'_1\right) \quad \log\left(\bar{k}^\alpha + (1-\delta)\bar{k} - k'_2\right) \quad \cdots \quad \log\left(\bar{k}^\alpha + (1-\delta)\bar{k} - k'_{\text{nkk}}\right) \right] \right\}$$

We fixed the state variable to take the value $\bar{k}$ but we can naturally do this for all values in the grid. Ultimately, this allows us to fill in all elements of the $1 \times$ `nkk` vector V_1. If we reached the fixed point in the very first iteration, then V_1 and our initial guess V_0 will be almost identical. We can verify this through different ways. Usually we simply take the Euclidean norm of $(V_1 - V_0)$ divided by that of V_0. A reasonable alternative would be to compute the absolute value of the largest element of $(V_1 - V_0)$. If either of these metrics is sufficiently small then we know that the fixed point has been found.

Since convergence is hardly achieved after a single iteration, suppose this is not the case. More specifically, say that the ratio of Euclidean norms (our chosen measure of distance) is greater than an arbitrarily small tolerance parameter ε. We now use V_1 as our initial guess for the value function and repeat the procedure all over again. As we cannot ensure that V_1 is zero across all grid points, the second iteration does the following:

$$V_2(\bar{k}) = \max \left\{ \begin{bmatrix} \log\left(\bar{k}^{\alpha} + (1-\delta)\bar{k} - k_1'\right) + \beta V_1(k_1') \\ \log\left(\bar{k}^{\alpha} + (1-\delta)\bar{k} - k_2'\right) + \beta V_1(k_2') \\ \vdots \\ \log\left(\bar{k}^{\alpha} + (1-\delta)\bar{k} - k_{\texttt{nkk}}'\right) + \beta V_1(k_{\texttt{nkk}}') \end{bmatrix}^{\mathrm{T}} \right\}$$

As we loop across all $\texttt{nkk}$ values the state variable can take we eventually populate the $1 \times \texttt{nkk}$ vector V_2. We take again the distance between the updated value function and the guess, V_2 and V_1, respectively, and see if that metric is still greater than the tolerance parameter ε.

In conclusion, the value function algorithm simply does the following. In iteration $i \geq 0$ we have a value function: $V_i(k)$ is a $1 \times \texttt{nkk}$ vector whose elements are all known. In that stage of the algorithm, we compute the following for every possible level of state capital:

$$V_{i+1}(k) = \max \left\{ \begin{bmatrix} \log\left(k^{\alpha} + (1-\delta)k - k_1'\right) + \beta V_i(k_1') \\ \log\left(k^{\alpha} + (1-\delta)k - k_2'\right) + \beta V_i(k_2') \\ \vdots \\ \log\left(k^{\alpha} + (1-\delta)k - k_{\texttt{nkk}}'\right) + \beta V_i(k_{\texttt{nkk}}') \end{bmatrix}^{\mathrm{T}} \right\}, \qquad \forall\, k = \{k_1, k_2, ..., k_{\texttt{nkk}}\}$$

The CMT ensures that there exists an iteration i for which $V_{i+1} = V_i = V^*$, such that applying the operator to V^* gives back V^* itself. In the algorithm we have built this means that V_i and V_{i+1} are sufficiently similar for us to declare that is indeed the unique fixed point. A more formal definition of *sufficiently similar* can entail the following rule:

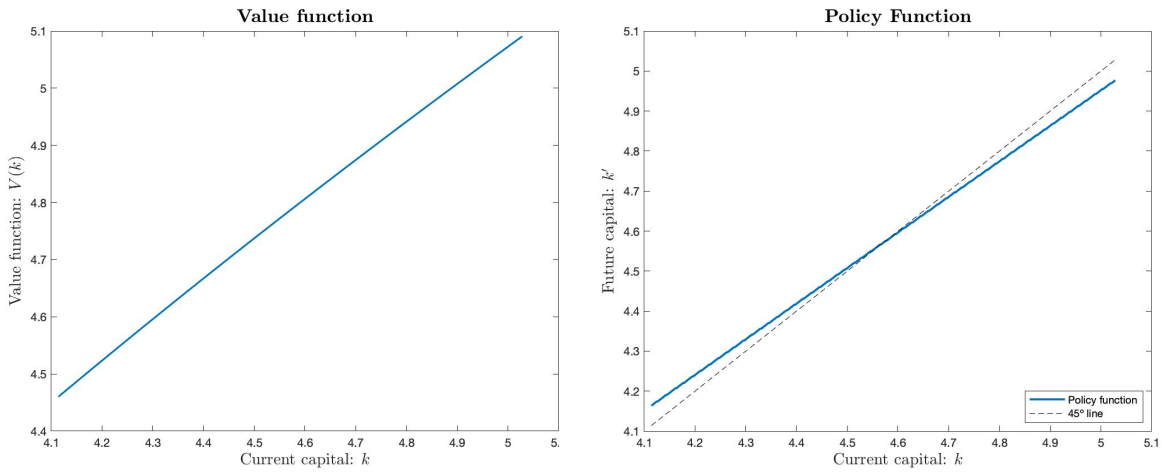
$$\begin{cases} \frac{||V_{i+1} - V_i||}{||V_i||} \geq \varepsilon & \implies \text{set } V_i = V_{i+1} \text{ and start all over} \\ \frac{||V_{i+1} - V_i||}{||V_i||} < \varepsilon & \implies \text{fixed point has been reached} \end{cases}, \tag{10}$$

where $||\cdot||$ denotes the Euclidean norm of a vector, and ε is the arbitrarily small tolerance parameter. Once all $\texttt{nkk}$ elements of V_{i+1} are found we can then evaluate whether the fixed point has been reached according to (10). Additionally, notice that in each iteration of this procedure we can also keep track of the best choice of capital, k', for each possible level of k. This is how we can ultimately obtain the policy function.

In Figure 3 below we plot the output produced by this algorithm given a certain parametrization of the model. The right panel is the solution the planner wanted to find: a policy function which tells him how

many units of capital should be saved today (k', on the vertical axis) for every possible value of capital he may now have (k, on the horizontal axis). The value function is shown on the right panel. It maps each possible state level of capital in the grid to the discounted utility level of optimally solving the problem.

Figure 3: Solution to the neoclassical growth model without uncertainty



Notes: Solution to the model obtained after 88 iterations and 0.23 seconds, with a convergence parameter of $\varepsilon = 10^{-5}$. The model was solved with an equally spaced capital grid of $\texttt{nkk} = 500$ points and the following parametrization: $\alpha = 0.36$, $\beta = 0.9$, and $\delta = 0.025$.

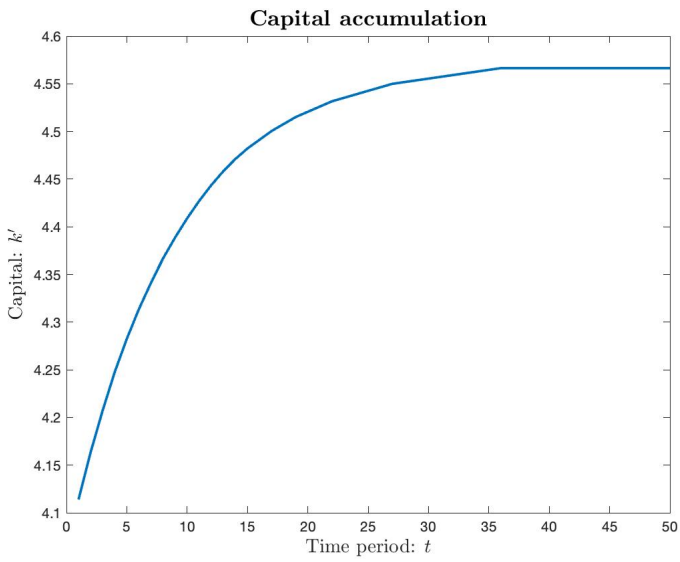
The policy function is particularly useful for us to understand how an actual economy would behave if governed by (9). Despite its deterministic setting, we can randomly draw an initial level of capital and use the policy function to track the optimal sequence of capital chosen by a planner over a finite period of time. Say, $T = 50$ periods. Figure 4 below plots that time series evolution starting with the lowest possible value of capital in the grid.

4 Stochastic dynamic programming problems

In the previous section we solved up a deterministic version of the neoclassical growth model with a specific numerical method: value function iteration. Since that version did not include any source of stochasticity we properly labelled it as deterministic. This is hardly ever the case in economic problems. Indeed, there often exists some degree of uncertainty either at the individual or aggregate levels. We will now change that.

In the first subsection we will go back to the neoclassical growth model and introduce a common source of uncertainty: there are random productivity disturbances affecting output production. More specifically, we will now consider the following production function: $f(k, z) = e^z \cdot k^\alpha$, where z is a stationary AR(1) idiosyncratic productivity process whose innovations are normally distributed. Formally, we can describe

Figure 4: Simulation of the neoclassical growth model without uncertainty



said process by the law of motion $z_t = \rho\, z_{t-1} + \epsilon_t$, with serial correlation $|\rho| < 1$ and $\epsilon \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

An important assumption is that productivity is serially correlated, $\rho \neq 0$. This implies that today's value is not only informative for today's decision but also brings with it information about the likelihood of tomorrow's realization. This is very useful for the planner because, given his current optimal choices, he no longer knows *ex-ante* the value of solving the problem from tomorrow on — future values of productivity are yet unknown. He can, however, rely on the probability distribution brought about by the current value of z to compute conditional expectations, which he then takes into account.

This is also valid in the second subsection, where we further emphasize the point that value function iteration can be a powerful numerical tool even when we deal with more complicated problems. To do so we depart entirely from the neoclassical growth literature and rely instead on a model of firms' labour demand with costs of adjustment. Those labour choices entail both *intertemporal* and *intratemporal* decisions[6], and are subject to fall in inaction regions due to the non-convex nature of adjustment costs.

4.1 Neoclassical growth model with uncertainty

At the beginning of each period of time the social planner now knows two things: *i)* the value of capital he chose in the previous period and which is carried over to the present, and *ii)* the realization of the productivity level which can either enhance of shrink the level of output to be produced in that period. It is the joint knowledge of the **state vector** (k, z) which allows the planner to optimally choose the **control**

[6]This resembles the inclusion of an endogenous labour supply decision in the neoclassical growth model.

variable, k'. We can then formulate the model as follows:

$$\begin{aligned} V(k,z) &= \max_{k',c} \Big\{ \log(c) + \beta\, \mathbb{E}_{z'|z}\big(V(k',z')\big) \Big\}, \quad \text{subject to} \quad c + k' = e^z \cdot k^\alpha + (1-\delta)k_t \;\wedge\; c \geq 0 \\ &= \max_{k' \in \Gamma(k,z)} \Big\{ \log\left(e^z \cdot k^\alpha + (1-\delta)k - k'\right) + \beta \int V(k',z') \cdot \varphi(\mathrm{d}z'|z) \Big\} \end{aligned} \tag{11}$$

Several things are worth mentioning. First, we rewrote consumption as a function of both states and the choice variable through the new resource constraint. Second, the non-negativity constraint on consumption (and capital) requires that $k' \in [0,\, e^z \cdot k^\alpha + (1-\delta)k] \equiv \Gamma(k,z)$. This is the new feasible set for k', which again depends on the state vector. And third, for the conditionally expected continuation value we explicitly integrate the value function over the levels productivity can take tomorrow *given* the current realization. Since z is a serially correlated process, the distribution of values in $t+1$ depends on the value drawn in period t. For convenience, call that conditional probability distribution function $\varphi(z'|z) \in (0,1)$.

The problem is very similar to its deterministic counterpart. Although the state vector has been enlarged by the inclusion of a stochastic process for productivity, the main properties of the model remain unaltered. These include Blackwell sufficient conditions. Since monotonicity and discounting still hold, we know that (11) is a contraction with a unique fixed point. More importantly, we also know that we can inevitably reach that fixed point for any initial guess we may have for the value function. This basically means that value function iteration is again a viable numerical method to solve the model. Below we discuss some important details about the algorithm to be built and show some resulting objects of interest: the value and policy functions, as well as the performance of a simulated economy over a lengthy period of time.

4.1.1 Discretize the state variables: capital and productivity

The algorithm which implements the value function iteration requires that the state vector be discretized. In this case we need a grid for capital and another one for productivity.

Start with capital. As before, it is sensible to set up a sufficiently fine grid around the steady-state level of capital. The FOC and envelope condition yield the following:[7]

$$-\frac{1}{e^z \cdot k^\alpha + (1-\delta)k - k'} + \beta \int \frac{\alpha e^{z'} \cdot k'^\alpha + 1 - \delta}{e^{z'} \cdot k'^{\alpha-1} + (1-\delta)k' - k''} \cdot \varphi(\mathrm{d}z'|z) = 0$$

$$\text{In steady-state} \implies -\frac{1}{k^\alpha + (1-\delta)k - k} + \beta \frac{\alpha k^{\alpha-1} + 1 - \delta}{k^\alpha + (1-\delta)k - k} = 0 \iff k = \left(\frac{\alpha\,\beta}{1-\beta(1-\delta)}\right)^{\frac{1}{1-\alpha}}$$

Since the source of stochasticity is killed off in steady-state ($z = 0$), this corresponds to the same level we had obtained in the deterministic model. We can then set up a grid of `nkk` points around this value.

[7] Productivity levels are drawn from an unbounded set, $z \in \mathbb{R}$. From Leibniz rule this mean we can just take the derivative of the value function under the integral sign.

The discretization of productivity, however, is slightly more challenging. In addition to the grid of points that z can take we must also discretize the conditional probability density function. We do this via the Tauchen method (see Appendix B), which transforms a *continuous* AR(1) process into a *discrete* first-order Markov chain. This procedure can then offer two important objects: *i)* a grid of $\texttt{nzz}$ points for the stochastic state variable z, and *ii)* a probability transition matrix Π of dimension $\texttt{nzz} \times \texttt{nzz}$.

$$\Pi = \begin{bmatrix} \pi(z_1'|z_1) & \pi(z_2'|z_1) & \cdots & \pi(z_{\texttt{nzz}}'|z_1) \\ \pi(z_1'|z_2) & \pi(z_2'|z_2) & \cdots & \pi(z_{\texttt{nzz}}'|z_2) \\ \vdots & \vdots & \ddots & \vdots \\ \pi(z_1'|z_{\texttt{nzz}}) & \pi(z_2'|z_{\texttt{nzz}}) & \cdots & \pi(z_{\texttt{nzz}}'|z_{\texttt{nzz}}) \end{bmatrix}$$

A generic element (m, n) of this matrix indicates the probability that tomorrow's productivity will be z_n' given that today's level is z_m. This necessarily implies that each of the rows sums up to 1: $\sum_{n=1}^{\texttt{nzz}} \pi(z_n'|z_m) = 1$, for all $m = \{1, 2, ..., \texttt{nzz}\}$.

4.1.2 VFI algorithm

We can solve the model in (11) via value function iteration with grids for capital and productivity of $\texttt{nkk}$ and $\texttt{nzz}$ points, respectively. The same trade-off holds: with finer grids we have increasingly accurate solutions but the required computational power is also larger. In this particular case, for each of the $\texttt{nkk} \times \texttt{nzz}$ possible combinations of the state vector (k, z), we want to know what is the best value of k' amongst all $\texttt{nkk}$ possibilities in the capital grid.

The algorithm operates as follows. Say in iteration $i \geq 0$ we have a certain value function: $V_i(k, z)$ is a $\texttt{nkk} \times \texttt{nzz}$ matrix whose entries are all known.[8] In each iterative step i we can rely on the transition matrix

[8] We know that the iterative process kicks off at $i = 0$ with whatever guess we want, usually $V_0(k, z) = 0$.

Π to obtain the conditional expectation of the continuation value:

$$\mathbb{E}_{z'|z}\big(V_i(k',z')\big) = \Pi \cdot \big(V_i(k',z')\big)^{\mathrm{T}}$$

$$= \begin{bmatrix} \pi(z_1'|z_1) & \pi(z_2'|z_1) & \cdots & \pi(z_{\mathtt{nzz}}'|z_1) \\ \pi(z_1'|z_2) & \pi(z_2'|z_2) & \cdots & \pi(z_{\mathtt{nzz}}'|z_2) \\ \vdots & \vdots & \ddots & \vdots \\ \pi(z_1'|z_{\mathtt{nzz}}) & \pi(z_2'|z_{\mathtt{nzz}}) & \cdots & \pi(z_{\mathtt{nzz}}'|z_{\mathtt{nzz}}) \end{bmatrix} \cdot \begin{bmatrix} V(z_1',k_1') & V_i(z_1',k_2') & \cdots & V_i(z_1',k_{\mathtt{nkk}}') \\ V_i(z_2',k_1') & V_i(z_2',k_2') & \cdots & V_i(z_2',k_{\mathtt{nkk}}') \\ \vdots & \vdots & \ddots & \vdots \\ V_i(z_{\mathtt{nzz}}',k_1') & V_i(z_{\mathtt{nzz}}',k_2') & \cdots & V_i(z_{\mathtt{nzz}}',k_{\mathtt{nkk}}') \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_1)\cdot V_i(z_j',k_1') & \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_1)\cdot V_i(z_j',k_2') & \cdots & \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_1)\cdot V_i(z_j',k_{\mathtt{nkk}}') \\ \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_2)\cdot V_i(z_j',k_1') & \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_2)\cdot V_i(z_j',k_2') & \cdots & \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_2)\cdot V_i(z_j',k_{\mathtt{nkk}}') \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_{\mathtt{nzz}})\cdot V_i(z_j',k_1') & \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_{\mathtt{nzz}})\cdot V_i(z_j',k_2') & \cdots & \sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|z_{\mathtt{nzz}})\cdot V_i(z_j',k_{\mathtt{nkk}}') \end{bmatrix}$$

The element (m,n) of this matrix indicates the expected value of choosing k_n' given that *today's* productivity level is z_m. Under the usual guess for the initial value function ($V_0(k,z) = 0$ for all (k,z)) this is conveniently reduced to a $\mathtt{nzz} \times \mathtt{nkk}$ matrix of zeros. Notice that amongst the two state variables only z explicitly and directly affects the expected continuation value. This comes again from the serial correlation nature of productivity: the probability distribution of z' depends crucially on the level drawn in the present. Knowledge of z is not only informative for the choice of capital today but also works as a signal of future decisions. The planner will carry k' to the future but he does not know what productivity will be. He can, at best, use today's level to infer the probability distribution of its future realization.

We now have all the required information to make the best choices for capital. Fix a state vector $(\bar{k},\bar{z})$. For that duple there are $\mathtt{nkk}$ possible choices of k'. Which one is to be chosen? Naturally, the one which yields the greatest value in the following $1 \times \mathtt{nkk}$ vector:

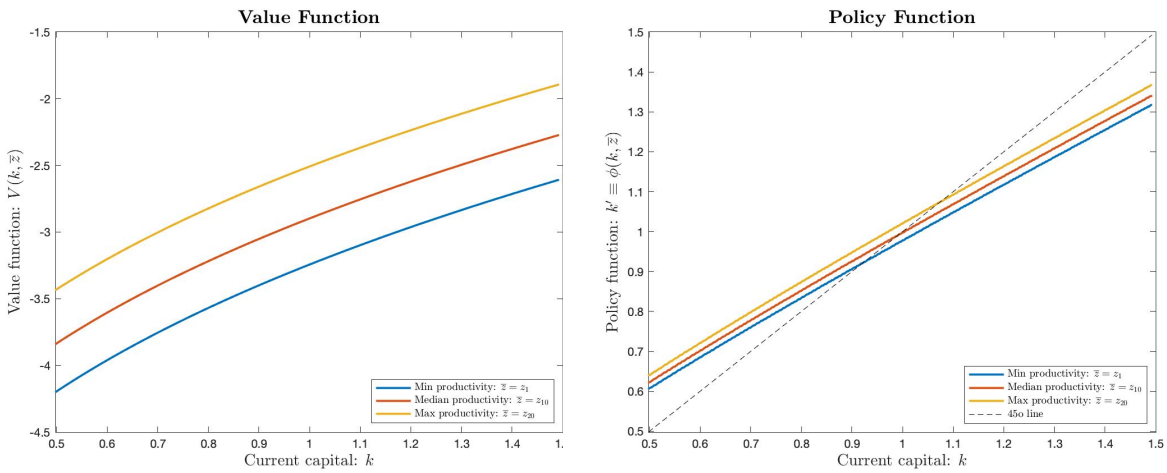
$$V_{i+1}(\bar{k},\bar{z}) = \max\left\{ \begin{bmatrix} \log\left(e^{\bar{z}}\cdot\bar{k}^{\alpha} + (1-\delta)\bar{k} - k_1'\right) + \beta\sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|\bar{z})\cdot V_i(z_j',k_1') \\ \log\left(e^{\bar{z}}\cdot\bar{k}^{\alpha} + (1-\delta)\bar{k} - k_2'\right) + \beta\sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|\bar{z})\cdot V_i(z_j',k_2') \\ \vdots \\ \log\left(e^{\bar{z}}\cdot\bar{k}^{\alpha} + (1-\delta)\bar{k} - k_{\mathtt{nkk}}'\right) + \beta\sum_{j=1}^{\mathtt{nzz}} \pi(z_j'|\bar{z})\cdot V_i(z_j',k_{\mathtt{nkk}}') \end{bmatrix}^{\mathrm{T}} \right\}$$

We can apply this max operator $\mathtt{nkk} \times \mathtt{nzz}$ times: one for each possible state vector (k,z). Ultimately, this will fill in all elements of matrix V_{i+1}. We can then evaluate whether we have reached the fixed point by measuring the distance between V_{i+1} and V_i. Notwithstanding other sensible alternatives, we can again rely on a distance measured by the Euclidean norm as displayed in (10).

Eventually, such an algorithm can produce the same two main objects of interest: a policy function for k' and the utility value the planner draws from optimally sticking to that rule. In the deterministic model

these were simpler functions as they were simple mappings from a one dimensional space to another. With stochastic productivity this is no longer the case. The value and policy functions now take two arguments as inputs, (k, z), for which they both spit out a scalar. For comparability we can, however, keep a certain level of productivity z constant and just vary the level of capital k. Figure 5 below shows this for a particular parametrization of the model.

Figure 5: Solution to the neoclassical growth model with uncertainty



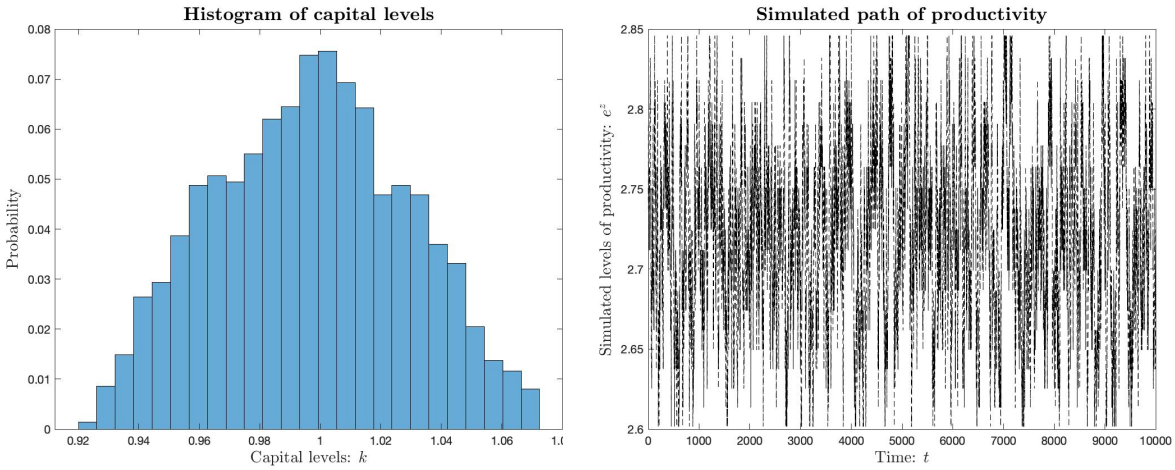
Notes: Solution to the model obtained after 91 iterations and 5.4 seconds, with a convergence parameter of $\varepsilon = 10^{-5}$. The model was solved with an equally spaced grid for capital of `nkk` $= 500$ points, a grid of `nzz` $= 20$ points for idiosyncratic productivity, and the following parametrization: $\alpha = 0.36$, $\beta = 0.9$, $\delta = 0.25$, $\rho = 0.95$, and $\sigma = 0.007$.

Suppose now we want to see how an economy like the one in (11) would actually behave. To do so, we can rely on a simulation exercise over a long period of time. Say, $T = 10\,000$ periods. As the Tauchen method gives us a transition matrix for productivity we can use it to simulate a path for productivity: $\{z_t\}_{t=1}^{T=10\,000}$. All we need is an initial level[9] and, from then on, we can just use the corresponding rows of the matrix Π to draw subsequent realizations of z. Ultimately, with an initial level of capital and the policy function for k' we can also track the optimal capital accumulation decisions over time.

Figure 6 below displays two outcomes of this simulation exercise. On the left-panel, the histogram of the capital levels at the end of the horizon. And on the right-panel the evolution of productivity throughout time.

[9] In the code attached to this section that initial level is drawn from the stationary distribution of z. Yet the importance of this initial condition vanishes as $T \to \infty$.

Figure 6: Simulation of the neoclaissical growth model with uncertainty



4.2 Labour demand model with adjustment costs

Value function iteration is a powerful method which allows to solve a large number of dynamic problems. In this final subsection we show that it can also be a useful tool when those dynamic problems are embedded with discontinuities and non-convexities, and when they are supplemented with static choices. To do this let us consider a particular variant of the models of firms' labour demand decisions with adjustment costs which can be found in chapter 9 of Adda and Cooper (2003).

Suppose in each period firms draw a random profitability level A, which is assumed to follow a stationary AR(1) process: $\log(A_t) = \rho \log(A_{t-1}) + \epsilon_t$, where $|\rho| < 1$ and $\epsilon \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$. Firms observe the realization of this random variable and also know the stock of workers they inherit from the previous period, n_{-1}. This duple (A, n_{-1}) is the state vector which firms rely on to make their optimal choices on *i)* employment (number of workers to be employed instantly that period, n), and *ii)* hours per worker (h). Formally, their dynamic programming problem can be expressed as follows:

$$V(A, n_{-1}) = \max_{n \geq 0, h \geq 0} \left\{ R(A, n, h) - \omega(n, h) - C(n_{-1}, n) + \beta \int V(A', n) \cdot \varphi(\mathrm{d}A' | A) \right\} \tag{12}$$

Firms generate their revenues according to their profitability level, A, and their effective labour usage (n employees, each working for h hours). More specifically, let the revenue function be $R(A, n, h) = A \cdot (nh)^{\alpha}$, where $\alpha \in (0, 1)$ is the curvature parameter. The chosen combination of n and h requires that compensation be paid in the amount of $\omega(n, h) = n \cdot (w_0 + w_1 h^{\eta})$, where w_0 is the base wage, w_1 the hourly rate, and η the constant elasticity between the hourly wage cost and hours per worker. This formulation assumes that the decision on average hours is static: firms merely trade off the marginal revenue associated with raising

average hours of work with the contemporaneous marginal wage cost of doing so. It is an *intratemporal* decision.

The choice of the number of workers, on the other hand, is an *intertemporal* one. This is justified by the existence of an adjustment cost function, $C(n_{-1}, n)$, which links current decisions with those of the subsequent period. We consider the following function:

$$C(n_{-1}, n) = \begin{cases} \Gamma_f + \gamma_f \cdot (n_{-1} - n) & \text{if } n_{-1} > n \\ 0 & \text{if } n_{-1} = n \\ \Gamma_h + \gamma_h \cdot (n - n_{-1}) & \text{if } n_{-1} < n \end{cases} \tag{13}$$

The subscripts f and h are meant to denote *firing* and *hiring*, respectively. Say the firm wants to sack people: they pay a fixed cost Γ_f and a linear cost for each of the $(n_{-1} - n)$ workers it wants to let go. The reasoning is the same for hiring decisions, albeit with potentially different parameters.

Finally, the firm discounts the continuation value by a factor $\beta \in (0, 1)$, and we again assume that the conditional probability distribution function associated with the stochastic state variable is $\varphi(A'|A)$.

The formulation displayed in (13) brings significant challenges to the model. First, $C(n_{-1}, n)$ is not an entirely differentiable function as it exhibits a kink when $n_{-1} = n$. And second, the inclusion of hiring and firing fixed costs introduces non-convexities to the problem. These would not appear in the first-order conditions with respect to n but firms do take them into account —large fixed costs, for instance, will certainly refrain firms from carrying out small adjustments. We will then have to rely on numerical methods to find a policy function for employment: $n \equiv \psi(A, n_{-1})$.

None of these issues applies to the decision on average hours of work. In fact, we can easily take a first-order condition which describes the optimal choice of h:

$$\text{FOC wrt } h: \quad \alpha A n^{\alpha} h^{\alpha-1} - \eta w_1 n h^{\eta-1} = 0 \iff h = \left(\frac{\eta w_1}{\alpha} \cdot \frac{n^{1-\alpha}}{A} \right)^{\frac{1}{\alpha-\eta}} \iff h = \theta \cdot A^{\frac{1}{\eta-\alpha}} \cdot n^{\frac{1-\alpha}{\alpha-\eta}}$$

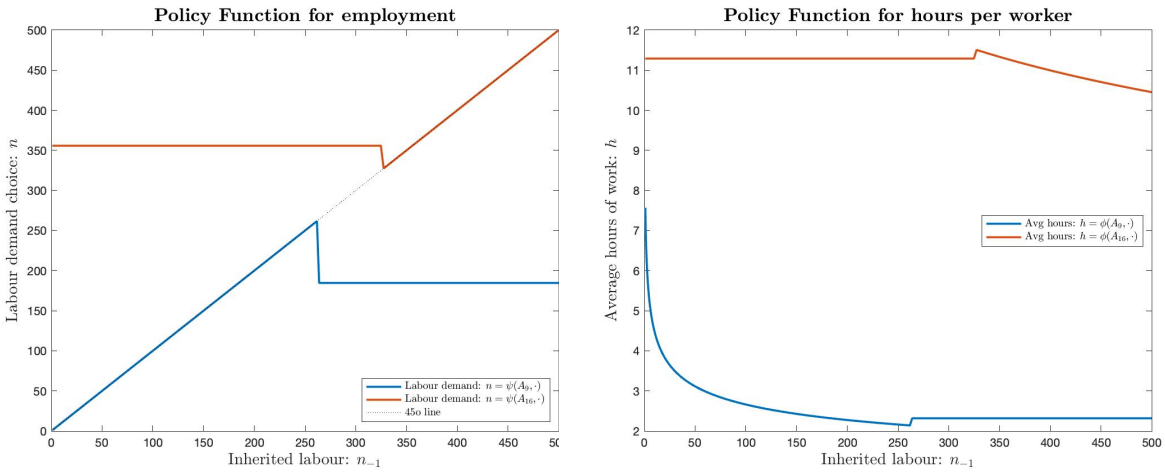
where $\theta = (\eta w_1/\alpha)^{1/(\alpha-\eta)}$ is a constant value defined by multiple parameters of the model. From the condition above it follows that the static decision about hours per worker depends on the current profitability level, A, and the optimal employment decision, n. But we also know that the latter is ultimately defined by the state vector:

$$h \equiv \phi(A, n_{-1}) = \theta \cdot A^{\frac{1}{\eta-\alpha}} \cdot \Big(\psi(A, n_{-1})\Big)^{\frac{1-\alpha}{\alpha-\eta}}$$

As long as we find a policy function for the dynamic problem, we can immediately obtain that of the static problem too. In order to do so we rely again on a value function iteration algorithm. Since equation (12) defines a contraction mapping we know that a sequence of value functions defined by that operator, and starting from whatever guess, will inevitably converge to a unique fixed point.

Using the procedure described in the previous subsection we can eventually obtain the following objects of interest:

Figure 7: Solution to the labour demand model with adjustment costs



***Notes*:** Solution to the model obtained after 84 iterations and 11.9 seconds, with a convergence parameter of $\varepsilon = 10^{-4}$. The model was solved with a logarithmic grid for labour of $\texttt{nnn} = 750$ points, a grid of $\texttt{naa} = 20$ points for profitability, and the following parametrization: $\alpha = 0.73$, $\beta = 0.93$, $\rho = 0.9$, $\sigma = 0.61$, $\Gamma_f = 2.9$, $\gamma_f = 1.34$, $\Gamma_h = 1.01$, $\gamma_h = 1.07$, $w_0 = 0.33$, $w_1 = 0.01$, and $\eta = 1.92$.

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A Proof of the Contraction Mapping Theorem (CMT)

The proof proceeds in three different stages. We first lay out some concepts which need to be used in the subsequent steps. Using these and the concepts of metric space and contraction we then show that a generic contraction T defines a convergent sequence. Under the assumption of a complete metric space this limit must belong to the space itself. Third and finally, we prove that this limit is in fact a unique fixed point.

We then start with some useful definitions. Notice the theorem relies on the existence of a *complete* metric space. The latter can be defined as follows.

Definition 2. ***(Complete metric space).*** *A metric space (S, d) is complete if every Cauchy sequence in S converges to an element in S.*

From the definition above, when every Cauchy sequence in S has a well-defined limit in S itself the metric space is said to be complete. Clearly this requires that we know what a Cauchy sequence is.

Definition 3. ***(Cauchy sequence).*** *A sequence $\{x_i\}_{i=0}^{\infty}$ in S has the Cauchy property if it is convergent. Formally, for each arbitrarily small $\epsilon > 0$ there exists a sequence index N_ϵ for which the following holds:*

$$d(x_j,\, x_i) < \epsilon, \quad \forall\, j, i \geq N_\epsilon$$

In other words, a sequence has the Cauchy property if it has a well-defined limit or, equivalently, if it is convergent. In a complete metric space (S, d) all such sequences converge to an element of S. Formally, this means that $\lim_{i\to\infty}\{x_i\} \in S$.

We will rely on these concepts for the second stage of the proof. Consider a contraction $T : S \to S$ with modulus $\beta \in (0, 1)$ in the metric space (S, d). By iteratively applying this operator over a generic element $x \in S$ we can define a sequence as follows:

$$Tx_i = x_{i+1}$$

The operator is applied over $x_i \in S$ and gives back $x_{i+1} \in S$. The sequence defined by the expression above can then be formally denoted by $\{x_i\}_{i=0}^{\infty}$. Together with the concept of a contraction (see **Definition 1**) this generic formulation implies the following about the first two elements:

$$d(x_1,\, x_2) = d(Tx_0,\, Tx_1) \leq \beta\, d(x_0,\, x_1),$$

where the first equality comes from the fact that $x_1 = Tx_0$ and $x_2 = Tx_1$, and the inequality is explained by the definition of a contraction. Let us now do the same to the next two elements of the sequence:

$$\begin{aligned} d(x_2,\, x_3) = d(Tx_1,\, Tx_2) \leq \beta\, d(x_1,\, x_2) = \beta\, \underbrace{d(Tx_0,\, Tx_1)}_{\leq \beta\, d(x_0, x_1)} \leq \beta^2 d(x_0, x_1) \\ \implies d(x_2, x_3) \leq \beta^2 d(x_0, x_1) \end{aligned} \tag{14}$$

We can obviously generalise the implication of (14) to the entire sequence defined by the contraction T. Indeed, for any integer $i = \{1, 2, ...\}$ the following is true:

$$d(x_i, x_{i+1}) \leq \beta^i d(x_0, x_1) \tag{15}$$

Equation (15) already shows that the distance between any two consecutive elements of the sequence than that between the initial two elements. This happens because $\beta^i \in (0, 1)$ for any integer $i = \{1, 2, ...\}$. But to show that the sequence is convergent we must demonstrate that the distance between any two elements (not necessarily consecutive ones) of the sequence becomes arbitrarily small after a finite number application of T. To prove this we can rely on the fact that d satisfies the triangle inequality and equation (15), amongst other trivial mathematical identities.

For any $j > i \geq 1$ the following must be true:

$$\begin{aligned}
&d(x_i, x_j) \leq d(x_{j-1}, x_j) + d(x_{j-1}, x_i) \\
&\implies d(x_i, x_j) \leq \underbrace{d(x_{j-1}, x_j)}_{\leq \beta^{j-1} d(x_0, x_1)} + \underbrace{d(x_{j-2}, x_{j-1})}_{\leq \beta^{j-2} d(x_0, x_1)} + \underbrace{d(x_{j-3}, x_{j-2})}_{\leq \beta^{j-3} d(x_0, x_1)} + (...) + \underbrace{d(x_i, x_{i+1})}_{\leq \beta^i d(x_0, x_1)} \\
&\implies d(x_i, x_j) \leq \beta^{j-1} d(x_0, x_1) + \beta^{j-2} d(x_0, x_1) + \beta^{j-3} d(x_0, x_1) + (...) + \beta^i d(x_0, x_1) \\
&\iff d(x_i, x_j) \leq \beta^i \cdot \left(\beta^{j-1-i} + \beta^{j-2-i} + \beta^{j-3-i} + (...) + \beta^{i-i}\right) \cdot d(x_0, x_1) = \beta^i \sum_{k=0}^{j-1-i} \beta^k \times d(x_0, x_1) \\
&\implies d(x_i, x_j) \leq \beta^i \sum_{k=0}^{\infty} \beta^k \times d(x_0, x_1) = \frac{\beta^i}{1-\beta} \times d(x_0, x_1) \\
&\implies d(x_i, x_j) \leq \frac{\beta^i}{1-\beta} \times d(x_0, x_1)
\end{aligned}$$

Focus on this last implication. First, it holds for any two elements of the sequence. Second, the right-hand side can be conveniently relabelled as ϵ. This means that for every $j > i \geq 1$ there exists indeed a value ϵ for which $d(x_i, x_j) \leq \epsilon$. Notice that we can also set up a strict inequality by removing an infinitesimally small $\varsigma > 0$ to ϵ.

In any case, the last implication shows an important conclusion: the sequence $\{x_i\}_{i=0}^{\infty}$ defined by the operator T is Cauchy because it is convergent. And since (S, d) is, by assumption, a complete metric space, then it follows that $\{x_i\}_{i=0}^{\infty}$ actually converges to another element of S: $\lim_{i\to\infty} x_i = x^* \in S$.

This brings us to the third and final stage of the proof. We must ultimately show that the limit x^* of the sequence defined by the operator is a unique fixed point of T. In other words, we want to prove that *i)* x^* is an existing fixed point, and *ii)* it is also unique.

Existence. We want to show that $x^* \in S$ is an existing fixed point of T, such that applying T over x^* yields back x^*. Formally: $Tx^* = x^*$. Take the definition of x^* as the limit of the sequence:

$$Tx^* = T \lim_{i\to\infty} x_i = \lim_{i\to\infty} Tx_i = \lim_{i\to\infty} x_{i+1} = x^*$$

We had already demonstrated that the sequence defined by the operator converges to $x^* \in S$. This now proves that the limit is actually a fixed point. Existence is proved.

Uniqueness. We prove the fixed point is unique through contradiction. Suppose then it is not. More specifically, say there are two fixed points: $Tx^* = x^*$ and $Ty^* = y^*$, with $x^* \neq y^*$. The distance between the two is given by $d(x^*, y^*) = \gamma > 0$. Using the definition of a contraction:

$$\gamma = d(x^*, y^*) = d(Tx^*, Ty^*) \leq \beta d(x^*, y^*) = \beta\gamma$$
$$\implies \gamma \leq \beta\gamma \iff \beta \geq 1,$$

which is a contradiction because we know that the modulus must lie between 0 and 1: $\beta \in (0, 1)$. The assumption that there are more than one fixed point cannot be sustained and uniqueness is thereby proved.

□

B The Tauchen method

The Tauchen method builds upon the contribution of Tauchen (1986). It basically allows us to approximate a *continuous* autoregressive process via a *discrete* Markov chain. Suppose we have stationary AR(1) process with normally distributed innovations:

$$z_t = \rho z_{t-1} + \epsilon_t, \quad \text{with} \quad |\rho| < 1 \text{ and } \epsilon \overset{iid}{\sim} \mathcal{N}(0, \sigma^2) \tag{16}$$

Stationarity is crucial to define the first two *unconditional* moments of the process: let $\mu \equiv \mathbb{E}(z) = 0$ and $\Sigma \equiv \text{Var}(z) = \sigma^2/(1-\rho^2)$.

These are moments of a continuous variable since ϵ is drawn precisely from a continuous (normal) distribution. Taking conditional expectations of z would then require us to compute the integral shown in equation (11). This is a heavy object: we are basically asked to weight the value function by the conditional probability of all possible realizations of z tomorrow. As there are infinitely many possible values in a continuous variable, we would have to somehow approximate this integral. Although there are indeed methods to do so, we will rely on a simpler alternative. Instead of working with a continuous z the Tauchen method discretizes it into a first-order Markov chain. Eventually, we should be able to obtain two objects:

1. A grid of `nzz` points for the stochastic variable z;

2. A (square) transition matrix Π of dimension $\texttt{nzz} \times \texttt{nzz}$.

Start with the grid. To set it up we must decide on how far apart should its lower and upper bounds be. This can be governed by the parameter $\kappa > 0$. More specifically, we let $-\kappa \times \sqrt{\Sigma}$ and $\kappa \times \sqrt{\Sigma}$ be the lower

and upper bounds of the grid, respectively, where $\sqrt{\Sigma}$ is the known standard deviation of the stochastic productivity variable. With Matlab's `linspace` command we can quickly set up an equally spaced grid of $\texttt{nzz}$ points, where $\Delta \equiv z_j - z_{j-1}$, with $j = \{2, ..., \texttt{nzz}\}$, corresponds to the distance between any two points of that grid.

This first procedure discretizes stochastic productivity, the *exogenous* state variable of the neoclassical growth model. In a first-order Markov chain that exogenous evolution is governed by a transition matrix, which is the second object we retrieve from the Tauchen method. Basically, we want a matrix which tells the probability of reaching z_n tomorrow, where $n = \{1, ..., \texttt{nzz}\}$ refers to a position in the newly created productivity grid, given that today's value is z_m.

The transition matrix can then be populated as follows. Start by fixing any z_m drawn today. The conditional probability of drawing any productivity value z'_n other than the min or the max is given the following way:

$$\begin{aligned}\pi(z'_n|z_m) &= \text{Prob}\left(z'_n - \frac{\Delta}{2} \leq \rho z_m + \epsilon \leq z'_n + \frac{\Delta}{2}\right) = \text{Prob}\left(\frac{z'_n - \frac{\Delta}{2} - \rho z_m}{\sigma} \leq \frac{\epsilon}{\sigma} \leq \frac{z'_n + \frac{\Delta}{2} - \rho z_m}{\sigma}\right) \\ &= \Phi\left(\frac{z'_n + \frac{\Delta}{2} - \rho z_m}{\sigma}\right) - \Phi\left(\frac{z'_n - \frac{\Delta}{2} - \rho z_m}{\sigma}\right), \quad \forall\, n = \{2, ..., \texttt{nzz} - 1\} \text{ and } \forall\, m = \{1, ..., \texttt{nzz}\}\end{aligned} \tag{17}$$

We know that ϵ is a normally distributed random variable with mean 0 and standard deviation σ. Therefore, $(\epsilon - 0)/\sigma$ must follow a standard normal distribution, whose cumulative distribution function is denoted by $\Phi(\cdot)$. To better understand what the approximation the Tauchen method does consider a productivity grid with infinitely many points: $\texttt{nzz} \to \infty$. In this extreme case the distance between the equally spaced points would then converge to 0: $\Delta \to 0$. But then notice that (17) actually becomes $\pi(z'_m|z_m) = \text{Prob}(z'_n \leq \rho z_m + \epsilon \leq z'_n) = \text{Prob}(\epsilon = z'_n - \rho z_m)$.

Condition (17) above allows us to fill in all intermediate columns (*i.e.* all those except the first and the last) of the transition matrix. We now consider the remaining two cases. First, for any z_m drawn today, the conditional probability of drawing the lowest possible value of productivity tomorrow (z'_1) is given by:

$$\pi(z'_1|z_m) = \text{Prob}\left(\rho z_m + \epsilon \leq z'_1 + \frac{\Delta}{2}\right) = \text{Prob}\left(\frac{\epsilon}{\sigma} \leq \frac{z'_1 + \frac{\Delta}{2} - \rho z_m}{\sigma}\right) = \Phi\left(\frac{z'_1 + \frac{\Delta}{2} - \rho z_m}{\sigma}\right) \tag{18}$$

And second, still for any z_m drawn today, the conditional probability of drawing the greatest possible value of productivity tomorrow ($z'_{\texttt{nzz}}$) is as follows:

$$\pi(z'_{\texttt{nzz}}|z_m) = \text{Prob}\left(z'_{\texttt{nzz}} - \frac{\Delta}{2} \leq \rho z_m + \epsilon\right) = \text{Prob}\left(\frac{\epsilon}{\sigma} \geq \frac{z'_{\texttt{nzz}} - \frac{\Delta}{2} - \rho z_m}{\sigma}\right) = 1 - \Phi\left(\frac{z'_{\texttt{nzz}} - \frac{\Delta}{2} - \rho z_m}{\sigma}\right) \tag{19}$$

Ultimately, this approximation offered by the Tauchen methods spits out a $\texttt{nzz} \times \texttt{nzz}$ transition matrix, whose rows indicate current values of productivity and columns stand for future values. Naturally this implies that each row sums up to 1: $\sum_{n=1}^{\texttt{nzz}} \pi(z'_n|z_m) = 1$, for any $m = \{1, ..., \texttt{nzz}\}$.

$$\Pi = \begin{bmatrix} \pi(z_1'|z_1) & \pi(z_2'|z_1) & \cdots & \pi(z_{\mathtt{nzz}}'|z_1) \\ \pi(z_1'|z_2) & \pi(z_2'|z_2) & \cdots & \pi(z_{\mathtt{nzz}}'|z_2) \\ \vdots & \vdots & \ddots & \vdots \\ \pi(z_1'|z_{\mathtt{nzz}}) & \pi(z_2'|z_{\mathtt{nzz}}) & \cdots & \pi(z_{\mathtt{nzz}}'|z_{\mathtt{nzz}}) \end{bmatrix}$$

$$= \begin{bmatrix} \Phi\left(\frac{z_1'+\frac{\Delta}{2}-\rho z_1}{\sigma}\right) & \Phi\left(\frac{z_2'-\frac{\Delta}{2}-\rho z_1}{\sigma}\right) - \Phi\left(\frac{z_2'+\frac{\Delta}{2}-\rho z_1}{\sigma}\right) & \cdots & 1-\Phi\left(\frac{z_{\mathtt{nzz}}'-\frac{\Delta}{2}-\rho z_1}{\sigma}\right) \\ \Phi\left(\frac{z_1'+\frac{\Delta}{2}-\rho z_2}{\sigma}\right) & \Phi\left(\frac{z_2'-\frac{\Delta}{2}-\rho z_2}{\sigma}\right) - \Phi\left(\frac{z_2'+\frac{\Delta}{2}-\rho z_2}{\sigma}\right) & \cdots & 1-\Phi\left(\frac{z_{\mathtt{nzz}}'-\frac{\Delta}{2}-\rho z_2}{\sigma}\right) \\ \vdots & \vdots & \ddots & \vdots \\ \Phi\left(\frac{z_1'+\frac{\Delta}{2}-\rho z_{\mathtt{nzz}}}{\sigma}\right) & \Phi\left(\frac{z_2'-\frac{\Delta}{2}-\rho z_{\mathtt{nzz}}}{\sigma}\right) - \Phi\left(\frac{z_2'+\frac{\Delta}{2}-\rho z_{\mathtt{nzz}}}{\sigma}\right) & \cdots & 1-\Phi\left(\frac{z_{\mathtt{nzz}}'-\frac{\Delta}{2}-\rho z_{\mathtt{nzz}}}{\sigma}\right) \end{bmatrix}$$

Finally, as productivity is approximated via a discrete first-order Markov Chain there is an additional object which may be of interest. Let ψ be the stationary distribution associated with this Markov chain. As long as irreducibility holds[10], the Perron-Frobenius theorem ensures that such distribution exists, it is unique and can be formally defined as follows:

$$\psi \cdot \Pi = \psi \iff \psi \cdot \left(\Pi - 1 \cdot \mathbf{I}\right) = 0, \tag{20}$$

where $\mathbf{I}$ is the identity matrix. The stationary distribution ψ is a $1 \times \mathtt{nzz}$ vector of probabilities. More specifically, the $j-$th element of ψ stands for the unconditional probability of reaching state z_j once we simulate the Markov chain a sufficiently large number of times.

From equation (20) it becomes clear that we can obtain that vector in two different ways. On the one hand, we can create an algorithm which computes $\psi_i \cdot \Pi = \psi_{i+1}$ and evaluates how similar ψ_i and ψ_{i+1} are. If they are close enough up to a tolerance parameter ε, then we can stop knowing the stationary distribution has been found. Otherwise, we set $\psi_i = \psi_{i+1}$ and start again. This is the chosen alternative in the Matlab code which implements the Tauchen method. On the other, notice that ψ is also defined as the left eigenvector of Π associated with the unit eigenvalue.

[10] A Markov chain is said to be irreducible whenever we can move from any one state to another within a finite number of steps.